Module 4: Supervised Learning
Submodule: 4.5 Softmax Regression
Lecture: Multiclass Classification with Softmax Regression

XCS229 Machine Learning

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Contents

I. Introduction

In this Lecture, we will introduce the concept of **multiclass softmax regression**, discussing its utility in classification tasks. We will also explain the basic idea of one-hot encoding.

II. Multiclass Softmax

The **multiclass softmax regression** allows for handling multiple discrete outcomes, which can be modeled using a multinomial distribution.

The multinomial distribution expresses probabilities across multiple classes. For instance, in a task where you need to classify images as {'cat', 'dog', 'car', 'bus'} we define $K = 4$ corresponding to these classes.

One-Hot vector

One-hot vector is a crucial concept in transforming categorical data into a numerical format suitable for machine learning. In this representation, each class is represented as a binary vector of length K (i.e., $y \in \{0,1\}^k$).

Examples, in $K = 4$:

- For 'cat': $\mathbf{y}_{\text{cat}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ is class 1.
- For 'dog': $\mathbf{y}_{\text{dog}} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ is class 2.
- For 'car': $\mathbf{y}_{\text{car}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is class 3.
- For 'bus': $\mathbf{y}_{\text{bus}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ is class 4.

The one-hot encoding ensures that there is only a single '1' present in any vector, which indicates the true class.

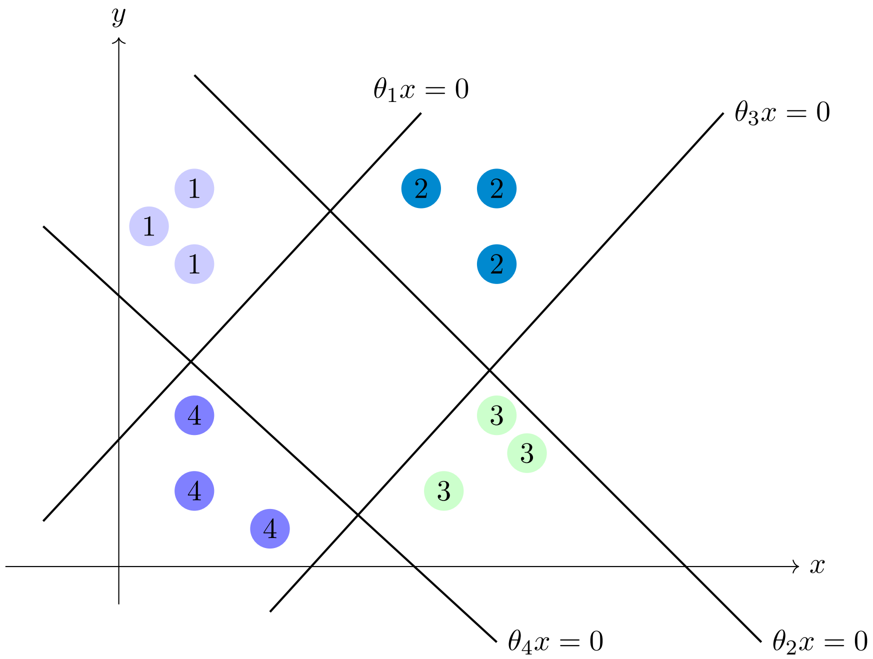
III. Multiclass Classification

In this section, we will explore how to achieve multiclass classification by utilizing hyperplanes and the concept of softmax.

To understand how multiclass classification works, let's consider an example dataset, represented as clusters of points in a feature space. We can visualize the data points and label them according to their respective classes. The colors used to distinguish classes are merely for visual assistance. In multiclass classification, our primary objective is to find hyperplanes that separate each class from the others.

Defining Hyperplanes

The concept of hyperplanes can be visually represented in a two-dimensional space. The equations of the hyperplanes that separate classes will provide a good visual understanding. The diagram below illustrates this idea:



The diagram illustrates different classes of data points represented in a two-dimensional feature space. The colored regions are indicative of the clusters, while the hyperplanes separate these classes.

For each hyperplane, we utilize parameters defined as:

$$\theta_i \cdot x = 0$$

This decomposition helps us visualize the separation between classes in the given feature space.

Softmax Function

Now we will transform the computed values into a probability distribution using the **softmax function**. The general format is given by:

$$P(y = k) = \frac{\exp(\theta_k \cdot x)}{\sum_{j=1}^{k} \exp(\theta_j \cdot x)}$$

The probabilities are normalized to sum to 1, allowing us to classify the input points based on their computed probabilities. E.g:

$$\begin{array}{cccccc}
\theta_1 x = 0.7 & & e^{0.7} & & {}^{e^{0.7}}/_{e^{0.7}} + e^{-0.5} + e^{-0.5} + e^{-0.2} \approx 46\% \\
\theta_2 x = -0.5 & \text{convert} & e^{-0.5} & & {}^{e^{-0.5}}/_{e^{0.7}} + e^{-0.5} + e^{-0.5} + e^{-0.2} \approx 14\% \\
\theta_3 x = -0.1 & \Longrightarrow & e^{-0.1} & \Longrightarrow & {}^{e^{-0.1}}/_{e^{0.7}} + e^{-0.5} + e^{-0.5} + e^{-0.2} \approx 21\% \\
\theta_4 x = -0.2 & & e^{-0.2} & & {}^{e^{-0.2}}/_{e^{0.7}} + e^{-0.5} + e^{-0.5} + e^{-0.2} \approx 19\%
\end{array}$$

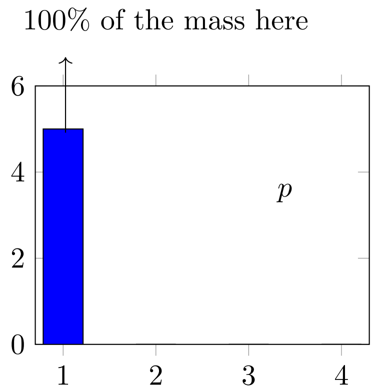
We have covered the fundamentals of multiclass classification, focusing on how to represent data, compute relevant values, and convert them into a probability distribution using the softmax function. Understanding these concepts is essential for implementing effective classification models in machine learning.

IV. Cross Entropy

In this section, we will discuss training models for multi-class classification. We will cover one-hot labels to represent true category distributions, how we estimate probabilities for each class during training, the concept of cross-entropy to quantify the difference between true and estimated distributions, and the role of the softmax function in converting raw outputs into interpretable probabilities.

- **Training Models:**

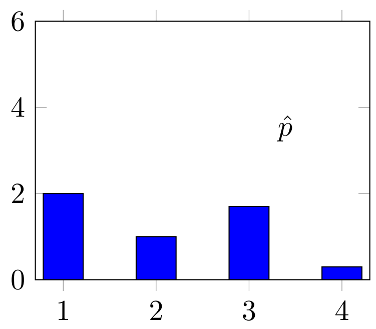
  When training models, we begin with one-hot labels indicating which class an instance belongs to. For example, if we are working with four classes, our true probability distribution can be represented as follows:



  In this diagram, p represents our true probability distribution where all the mass is on class one, indicating 100% confidence in that class.

- **Estimating Probabilities:**

  While training, our model will estimate probabilities for the classes, but it will often not be completely confident. This means we might have a distribution like the following:



In this second diagram, $\hat{p}$ represents our estimated probability distribution for the classes, capturing some uncertainty in the predictions.

Cross Entropy and Softmax

To train our models, we minimize the **cross-entropy loss** defined as:

$$C(p, \hat{p}) = -\sum_{y=1}^{k} p(y) \log(\hat{p}(y))$$

This loss function reflects the difference between the true probability distribution (p) and the estimated distribution ($\hat{p}$). By minimizing the cross-entropy, we are aligning our estimated probabilities with the true distribution.

When we have one-hot encoded p, only one term in the summation survives for the true class. If $p(y = k)$, we get:

$$C(p, \hat{p}) = -\log(\hat{p}(y_k))$$

This shows that minimizing the cross-entropy is equivalent to maximizing the likelihood of the true class under our model, which leads to effective learning.

Both minimizing cross-entropy and applying the softmax function yield the same results in the context of multiclass classification, specifically when dealing with categorical outcomes.

The process of training models involves estimating class probabilities based on one-hot labels and minimizing the cross-entropy loss. The softmax function plays an essential role in translating raw model outputs into meaningful probability distributions. Understanding these concepts is crucial for effective machine learning model training.

V. Summary

In this lecture on Multiclass Classification with Softmax Regression, we have covered the following key concepts:

- **Multiclass Softmax Regression:** This approach is used for handling multiple discrete outcomes and can be modeled using a multinomial distribution.
- **One-Hot Encoding:** A technique to represent categorical outcomes as binary vectors. Each class is represented by a unique vector containing a single '1' and the rest '0's, making it suitable for machine learning algorithms.
- **Hyperplanes:** In multiclass classification, hyperplanes are used to separate different classes in the feature space, providing a geometric interpretation of class boundaries.
- **Softmax Function:** This function transforms the output scores from the model into a probability distribution across the classes, ensuring that probabilities sum to 1.
- **Cross-Entropy Loss:** A crucial loss function for training models, measuring the discrepancy between the true distribution (one-hot encoded) and the estimated probabilities generated from the softmax function.
- **Probabilities Estimation:** The model estimates probabilities for each class, which may not always show complete confidence, especially in cases of uncertainty.
- **Connection of Concepts:** The interplay between one-hot encoding, softmax, and cross-entropy is essential for effective training of multiclass classification models in machine learning.